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Influence of Teachers' Demographic Characteristics on Technology Integration Readiness in Secondary Schools in Olorunda Local Government Area, Osun State

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Abstract

The integration of technology in education has become increasingly important in enhancing teaching effectiveness and improving students' learning outcomes. Despite this, teachers' readiness remains a major concern, particularly in developing countries. This study examined the influence of teachers' demographic characteristics on their readiness to integrate technology into the teaching-learning process in secondary schools. Specifically, the research investigated teachers' readiness to integrate technology and assessed whether significant differences exist by gender, age, and years of teaching experience. The descriptive survey research design was employed. From schools in Olorunda Local Government of Osun State, Nigeria, a sample of 150 secondary school teachers was chosen at random. A structured questionnaire called Teachers' Readiness to Integrate Technology in the Teaching-Learning Process was used to gather data. Pearson Product-Moment Correlation was performed to establish the dependability of the instrument, which provided a reliability coefficient of 0.82. Data were analysed using descriptive statistics such as frequency

counts, percentages, Means, and Standard Deviation, while inferential statistics, including t-test and Analysis of Variance (ANOVA), were employed to evaluate the hypotheses. The study revealed that secondary school teachers are ready to integrate technology into the teaching-learning process. The study also revealed that there are significant differences in teachers' readiness to integrate technology in the teaching-learning process based on age, gender and years of experience. Based on the findings, it is recommended that all categories of teachers should be exposed to ICT training to improve their skills and willingness to use technology in their teaching-learning processes, irrespective of their ages, gender and years of experience.

Keywords: Teachers' readiness, demographic characteristics, Technology, Teaching-learning process

Introduction

In modern educational systems, technology has emerged as one of the most important tools for teaching and learning. Technology is the application of scientific knowledge, techniques, systems, and inventions to increase human capabilities, solve issues, and boost productivity across various sectors. It includes both digital and physical technologies for communication, information processing, and socioeconomic growth (Aondover, Oyeleye & Aliyu, 2024). In other words, technology offers new capabilities that result in profound organisational change. It enhances communication processes and presents innovative ways for disseminating information. Thus, information technology is a type of communication technology that includes networks, hardware, and software for data capture, transmission, storage, retrieval, manipulation, and display (World Bank, 2020). Additionally, technology includes a broad range of digital tools and programmes intended to improve efficiency in daily tasks (OECD, 2021). These include digital platforms like social media, messaging apps, and interactive learning tools, as well as communication devices like computers, radios, televisions, and smartphones.

The use of educational technology in the classroom is now a major component of modern teaching and learning. As the world rapidly becomes more digital, particularly during the pandemic when everyone needed technology to support education, teachers were expected to adapt to these changes and successfully integrate digital tools into their pedagogy. Rapid technological advancements have brought about a profound change in education, altering both how students learn and how teachers interact with them (Joseph et al., 2021). However, to effectively manage this shift, teachers must have access to professional development opportunities (Hennessy, D'Angelo, McIntyre, & Koomar, 2022).

Technological tools such as television and radio, computers, printers, scanners, software programs, projectors, interactive boards and the Internet, have been widely recognised as powerful tools capable of facilitating educational change and reform (Hermanto YB, & Srimulyani V.A., 2021). When effectively utilised, technology can expand access to education, enhance its relevance to the demands of the digital workplace, and improve the quality of teaching and learning. In

addition, technology helps make learning more engaging and interactive, enabling students to participate actively in the learning process. Through technological advancement, significant improvements have been recorded in the ways teachers communicate knowledge and perform their instructional responsibilities. Technology has therefore provided teachers with opportunities to move beyond the traditional chalk-and-talk approach to modern, innovative teaching methods.

The use of technology in education involves not only pedagogical changes but also changes in the classroom environment. For teachers to utilise technology effectively, they require adequate training, technical support, and opportunities to practise the use of technological tools. The integration of technology in education, therefore, requires teachers to develop both technical competence and pedagogical skills to use technology meaningfully in classroom instruction. Many educators face challenges, such as inadequate training, negative attitudes toward the use of technology in the classroom, and its underutilisation remains an issue. In educational settings, these barriers often prevent technology from realising its full potential. To help teachers smoothly incorporate technology into their teaching-learning, professional development programme designs should be designed and implemented for teachers (Sancar, Atal, & Deryakulu, 2021).

The use of technological tools and resources in the teaching and learning process is referred to as technology integration in education. It entails using technology in the curriculum through a range of tools that improve students' participation in worthwhile learning activities. The study by Zulueta et al. (2021) shows that teachers could include ICT and transform their old methods of education into a dynamic and creative one that is suitable for today's tech-savvy students. As a result, it creates a classroom that emphasizes learning and transforms the teacher into a facilitator and innovator. Technology is a multidimensional tool for modern instruction, from fostering engagement and participation to enhancing understanding, differentiation, and critical thinking. Given the high levels of perceived impact, the emphasis should now be on sustaining this momentum through ongoing training, group projects, and fair access to resources and tools so that all teachers and learners can profit from technology-enhanced instruction. It also helps reduce teachers' workload and improve student-teacher relationships.

According to Qureshi, Khan, Raza, Imran, and Ismail (2021), teachers' effectiveness fosters a more innovative school culture because they feel emboldened to discuss their achievements and difficulties with integrating technology. Despite the benefits, the effective integration of technology in teaching and learning may be influenced by several factors, including teachers' demographic characteristics such as age, gender, teaching experience, and professional qualifications. These characteristics can influence teachers' attitudes towards technology, their competence in using technological tools, and their willingness to adopt new teaching methods.

Teachers' demographic characteristics refer to the personal attributes that distinguish one teacher from another. Merriam-Webster Dictionary (2016) defines demographic characteristics as the distinct features of a person; it can also be described as the peculiar attributes of individuals. Demographic characteristics also include physical and social attributes such as age, gender, marital status, education, occupation, and geographical location. Demography, therefore, focuses on the

size, structure, and characteristics of human populations and how these characteristics change over time.

One important predictor of teachers' level of proficiency is their views about the adoption of digital technology in the classroom (Masalimova et al., 2022). Teachers' opinions about using ICT in the classroom are influenced by a number of variables, such as their age, gender, and degree of teaching experience (Estaji & Zhaleh, 2022). It has been proposed that a teacher's attitude toward ICT integration in the classroom is related to their age group (Makhlouf & Bensafi, 2021). Masry-Herzallah (2025) found that gender differences in technological self-efficacy significantly influence teachers' adoption and effective use of digital technologies in teaching. Additionally, compared to their more seasoned counterparts, beginner teachers were shown to have a more favourable opinion of ICT use in the classroom (Belda-Medina & Clavo-Ferrer, 2022).

Age has been identified as an important factor influencing teachers' use of technology in the classroom. Younger teachers tend to have higher levels of confidence in their ability to utilise technology in the teaching-learning process compared with older teachers. Joseph et al. (2021) explained that as teachers' age or years of experience increased, their technology readiness tended to decrease. Older teachers may sometimes be reluctant to integrate technology into their teaching because they are more comfortable with conventional methods of instruction. Consequently, younger teachers who have been exposed to digital technologies during their training may develop more positive attitudes towards the use of technology in education.

Gender is another factor that has been widely discussed in relation to technology use. Research shows that teachers' professional opinions regarding technology integration differ significantly based on gender. According to a study on gender dimensions in emerging technologies by Ayite et al. (2022), technology presents both relevant and distracting situations. Also, Amponsah et al. (2024) opined that male teachers frequently feel more educated and confident while utilising technology. However, recent studies indicate that gender differences continue to influence teachers' attitudes toward technology integration. For instance, Masry-Herzallah (2025) found that female teachers are increasingly receptive to the use of technology and perceive it as a valuable tool for enhancing instructional effectiveness, particularly when supported by adequate training and technological self-efficacy. Similarly, the importance of digital education in preparing students for a technologically driven future is widely acknowledged by teachers across gender groups, as emphasised by UNESCO (2023), which highlights the role of technology in fostering 21st-century skills and improving learning outcomes.

Nevertheless, disparities in technology integration still exist, as male teachers often report higher levels of confidence and perceived competence in using digital tools. This is supported by Sofal et al. (2024), who found that male teachers demonstrated higher digital competencies compared to their female counterparts. Recent evidence suggests that such differences are not universal, as some studies indicate that both male and female teachers can equally engage in technology integration when provided with adequate support and access to resources (Falloon, 2020).

Years of teaching experience also play a significant role in teachers' adoption of technology in classroom instruction. Recent research suggests that less experienced teachers tend to demonstrate higher levels of technology use due to their exposure to digital tools during their training, while more experienced teachers may rely more on traditional instructional methods. This position is supported by Tondeur et al. (2021), who found that early-career teachers exhibit higher levels of digital competence compared to their more experienced counterparts. However, Momdjian et al. (2024) also found that a teacher's years of teaching experience did not significantly influence ICT skill level, although less experienced teachers were noted to have higher ICT competence.

The successful integration of technology into teaching and learning in both developing and developed countries, including Nigeria, largely depends on teachers, who are responsible for implementing the curriculum in the classroom. Teachers, therefore, play a critical role in determining whether technology will be effectively integrated into teaching practices. However, factors such as teachers' age, gender, years of experience, and professional qualifications may influence their readiness to adopt technological innovations in the classroom.

Statement of the Problem

Despite increased investments in educational technologies, the extent to which teachers are ready to integrate these tools into classroom practices remains unbalanced, particularly in developing contexts such as Nigeria. While previous studies have examined factors influencing technology integration, limited attention has been given to how teachers' demographic characteristics, such as age, gender, teaching experience, and educational qualification, contribute to their readiness for technology integration at the secondary school level. Furthermore, existing studies present inconsistent findings regarding the influence of these variables, thereby creating a need for context-specific empirical investigation. This study, therefore, seeks to examine the influence of teachers' demographic characteristics on their readiness to integrate technology in the teaching-learning process in Olorunda Local Government Area of Osun State.

Purpose of the Study

The general purpose of the study was to examine the influence of teachers' demographic characteristics on their readiness to integrate technology in the teaching-learning process.

Specifically, the study aimed to;

1. examine the level of readiness of secondary school teachers to integrate technology into teaching-learning process
2. determine the difference in teachers' readiness to integrate technology into the teaching-learning process based on gender
3. determine the difference in teachers' readiness to integrate technology into the teaching-learning process based on age.
4. determine the difference in teachers' readiness to integrate technology into the teaching-learning process based on years of experience.

From the objectives stated, one research question and three hypotheses are formulated.

Research Questions

1. What is the level of secondary school teachers' readiness to integrate technology into the teaching-learning process?

Hypotheses

- H₀₁: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on gender
- H₀₂: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on age
- H₀₃: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on years of experience

Methodology

The descriptive research design of the survey type was employed. The design allowed the researcher to gather information, summarise, present and interpret for the purpose of generalising the findings. The population of this study comprised all secondary school teachers in Olorunda Local Government Area of Osun State. A total of 150 respondents were sampled for the study. A simple random sampling technique was used to select 10 teachers from each of 15 randomly selected schools, made up of eight (8) public schools and seven (7) private secondary schools. The study employed a self-constructed questionnaire titled "Teachers Technology Integration for Teaching-Learning Process Questionnaire" (TTITLPQ). The questionnaire was divided into two sections. Section A contained items on teachers' personal information, and Section B consisted of 16 items on teachers' technology integration for teaching and learning. The four Likert scales of Strongly Agree (4), Agree (3), Disagree (2), Strongly Disagree (1) were used to collect responses from respondents. A mean score of 2.50 was used as the decision rule. Any item with a mean score of 2.50 and above was considered "Agreed", while mean scores below 2.50 were considered "Disagreed". The instrument was presented to the supervisor of the researcher for approval after being clearly designed. To confirm the validity of the instrument, ICT experts were engaged, and their suggestions and revisions were taken into consideration. The instrument was subjected to content validity through expert review, and a Content Validity Index of 0.82 was obtained, indicating good validity. Test-retest was used to assess the instruments' consistency. Twenty teachers who were not included in the study sample were given the questionnaire. The reliability coefficients of the instruments were determined by analysing the data using Pearson's Product-Moment Correlation (PPMC) after the instruments were given to the same respondents twice in two weeks. For the questionnaires, a reliability coefficient of 0.82 was obtained. The researcher personally administered the instrument by visiting the selected schools and distributing the questionnaire to the respondents. The instrument was retrieved immediately from the respondents

in order to maintain a high level of return. Data collected were analysed using descriptive statistics (frequency count, percentage, mean, standard deviation), T-test and Analysis of variance.

Results

Demographic Information

This section represents data analysis for the demographic information included in the study. These include gender, age and years of teaching experience of the respondents.

Table 1: Distribution of Teachers according to Gender, Age and Years of Teaching Experience

Gender	Frequency	Percentage (%)
Male	76	50.7
Female	74	49.3
Total	150	100
Age	Frequency	Percentage (%)
21-30 years	26	17.3
31-40 years	65	43.3
41-50 years	32	21.3
51-60 years	21	14.0
60 years and above	6	4.0
Total	150	100
Teaching Experience	Frequency	Percentage (%)
0-5 years	35	23.3
6-10 years	74	49.3
11-15 years	20	13.3
16-20 years	6	4.0
20 years and above	15	10.0
Total	150	100

The distribution of responders by age, gender, and years of teaching experience is shown in Table 1. The data indicate that 50.7% of respondents are males and 49.3% are females, indicating that both genders are well represented in the survey. According to the respondents' age distribution, 17.3% of them are between the ages of 21 and 30, and 43.3% are between the ages of 31 and 40. Also, 21.3% of respondents are between the ages of 41 and 50, while 14.0% of all respondents are between 51 and 60 years of age, 4.0% of respondents are above 60 years of age. This indicates that most teachers are between the ages of 31 and 40. In addition, 23.3% of respondents have taught for 0-5 years, 49.3% for 6-10 years, 13.3% for 11-15 years, 4.0% for 16-20 years, and 10.0% for 20 years or more. This indicates that a greater proportion of the participants are relatively new to the teaching profession.

Section B: Analysis of Research Questions and Interpretation of Results

Research Question One: What is the level of secondary school teachers' readiness to integrate technology into the teaching-learning process?

Table 2: Analysis of Secondary School Teachers' Readiness to Integrate Technology into the Teaching-Learning Process

S/N	Items	SA	A	D	SD	Mean	SD
1	I feel more comfortable using conventional methods of teaching than using technology in the classroom	45 30.0%	37 24.7%	47 31.3%	21 14.0%	2.71	1.046
2	I feel comfortable in developing my ICT skills	62 41.3%	64 42.7%	20 13.3%	4 2.7%	3.23	.778
3	I prefer to use technology resources in teaching-learning process	66 44.0%	53 35.3%	17 11.3%	14 9.3%	3.14	.956
4	The stress associated with the use of technology discourages me from embracing it	31 20.7%	32 21.3%	54 36.0%	33 22.0%	2.41	1.050
5	I adopt technology resources in the teaching-learning process	39 26.0%	61 40.7%	34 22.7%	16 10.7%	2.82	.942
6	Most times, I feel reluctant to adopt technology resources in the teaching-learning process	27 18.0%	36 24.0%	54 36.0%	33 22.0%	2.38	1.02
7	I feel comfortable not improving on my ability to utilise technology resources in teaching-learning process because I feel it is not necessary	23 15.3%	25 16.7%	56 37.7%	46 30.7%	2.17	1.03

8	I do not have an interest in adopting technology resources in teaching-learning process	24 16.0%	27 18.0%	45 30.0%	54 36.0%	2.14	1.081
9	I would like to use technology resources in communication during teaching-learning process	55 36.7%	51 34.0%	22 14.7%	22 14.7%	2.93	1.050
10	I don't like using technology resources in assessment and evaluation	23 15.3%	36 24.0%	53 35.3%	38 25.3%	2.29	1.014
11	Technology should be integrated in the teaching-learning process	73 48.7%	45 30.0%	28 18.7%	4 2.7%	3.25	.851
12	Integrating technology in teaching-learning process doesn't give room for effective learning	25 16.7%	44 29.3%	38 25.3%	43 28.7%	2.34	1.067
13	ICT skills acquisition should be integrated in teaching profession	62 41.3%	50 33.3%	22 14.7%	16 10.7%	3.05	.995
14	I feel integrating technology into the classroom as a waste of time	30 20.0%	38 25.3%	36 24.0%	46 30.7%	2.35	1.117
15	ICT skills acquisition should be embraced by teachers	69 46.0%	47 31.3%	20 13.3%	14 9.3%	3.14	.976
16	Utilisation of ICT resources should	81	51	11	7	3.37	.816

be encouraged in the school curriculum	54.0%	34.0%	7.3%	4.7%	
Total	735(30.6)	697(29.04)	557(23.2)	411(17.13)	2.56

Results

The data presented in Table 2 shows the level of secondary school teachers' readiness to integrate technology into the teaching-learning process. A mean score of 2.50 was used as the benchmark for decision-making; items with mean scores of 2.50 and above were regarded as "Agreed," while those below 2.50 were regarded as "Disagreed."

The results indicate that teachers generally demonstrate a moderate to high level of readiness to integrate technology into the teaching-learning process, as evidenced by the overall mean score of 2.56, which is above the criterion mean of 2.50.

Specifically, respondents agreed that they feel comfortable improving their ICT skills (Mean = 3.23), prefer the use of technology resources in teaching-learning (Mean = 3.14), and support the integration of technology into the teaching-learning process (Mean = 3.25). They also agreed that ICT skills acquisition should be integrated into the teaching profession (Mean = 3.05) and embraced by teachers (Mean = 3.14), while the utilisation of ICT resources should be encouraged in the school curriculum (Mean = 3.37).

Furthermore, respondents disagreed with negative statements such as the stress associated with technology discouraging its use (Mean = 2.41), reluctance to adopt technology (Mean = 2.38), lack of interest in using technology (Mean = 2.14), and the perception that integrating technology is a waste of time (Mean = 2.35). This indicates a generally negative disposition toward technology integration.

However, a notable exception is that respondents agreed with the statement that they feel more comfortable using conventional teaching methods than technology (Mean = 2.71), suggesting that while teachers are willing to adopt technology, they may still retain a preference for traditional instructional methods.

Based on the aggregate mean score of 2.56, which exceeds the benchmark of 2.50, it can be concluded that secondary school teachers in the study area are generally ready to integrate technology into the teaching-learning process. Overall, the findings suggest that teachers possess a reasonable level of readiness to integrate technology into the teaching-learning process, although certain attitudinal and behavioural factors may still limit full adoption.

Hypothesis One: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on gender.

An independent samples test (t-test) was performed to provide an answer to the first hypothesis.

Table 3: Summary of t-test Analysis on the Difference in Teachers' Readiness to Integrate Technology in the Teaching-Learning Process based on Gender.

Group	N	Mean	S.D	T	Df	Sig (2-tailed)	Decision
Male	76	3.38	.78826	1.987	148	.049	Significant
Female	74	3.11	.9000				

Difference between means significant at 5% alpha level; df = 148; Critical t-value = 1.96 (two-tailed test)

The gender-specific results are displayed in Table 3. The calculated mean and standard deviation values for male and female teachers are (3.38, 3.11) and (.78826, .9000), respectively. The outcome also showed that the probability value of 0.049 is less than the 5% (0.05) significance level and the computed t-value of 1.987 is higher than the critical t-value of 1.96 at the 5% level of significance. Teachers' readiness to integrate technology into the teaching-learning process varies significantly depending on their gender, as indicated by the estimated t-value being higher than the critical t-value ($t = 1.987$, $df = 148$, $p = .049 < 0.05$).

This suggests that there is a statistically significant difference in the readiness of male and female educators to use technology in the teaching-learning process. This indicates that, with respect to the teaching-learning process, female teachers are less integrated with ICT than their male colleagues. As a result, male educators are more likely to influence how ICT is integrated into secondary education during the teaching-learning process.

Hypothesis Two: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on age.

Table 4a: One-Way ANOVA result of variation on the difference in teachers' readiness to integrate technology in the teaching-learning process based on age

Source: Age	Sum of Squares	Df	Mean Square	F	Sig.	Decision
Between Groups	12.293	4	3.073	2.755*	.030	Significant
Within Groups	161.767	145	1.116			
Total	174.060	149				

The results of the age-based differences in teachers' readiness to integrate technology into the teaching-learning process are shown in Table 4a. The findings indicate that age significantly affects instructors' readiness to incorporate technology into the teaching-learning process ($F(4,145) = 2.755$, $p = .030 < 0.05$). This suggests that there are significant age-based differences in teachers' preparedness to use technology in the teaching-learning process.

On the other hand, the mean comparison indicates that younger teachers score higher on average when it comes to their readiness to use technology in the classroom.

The null hypothesis is therefore rejected.

Table 4b: Tukey HSD Post-hoc Analysis of Teachers' Readiness Based on Age

(I) Age Group	(J) Age Group	Mean Difference (I-J)	Std. Error	Sig.	Decision
21–30 years	31–40 years	0.158	0.134	.503	Not Significant
21–30 years	41–50 years	0.332*	0.140	.043	Significant
21–30 years	51–60 years	0.397*	0.147	.031	Significant
21–30 years	60 years and above	0.468*	0.156	.020	Significant
31–40 years	41–50 years	0.174	0.137	.398	Not Significant
31–40 years	51–60 years	0.239	0.143	.276	Not Significant
31–40 years	60 years and above	0.310	0.149	.045	Significant
41–50 years	51–60 years	0.065	0.140	.913	Not Significant
41–50 years	60 years and above	0.136	0.146	.658	Not Significant
51–60 years	60 years and above	0.071	0.151	.902	Not Significant

* Mean difference is significant at 0.05 level

To further determine the specific group differences among the age groups, a post-hoc analysis using Tukey HSD was conducted. The results, as presented in Table 4b, revealed that significant differences exist between younger teachers (21-30 years) and older age groups (41-50 years, 51-60 years, and 60 years and above). This indicates that younger teachers demonstrate higher readiness to integrate technology compared to older teachers.

Hypothesis Three: There is no significant difference in teachers' readiness to integrate technology into the teaching-learning process based on years of experience.

Table 5a: One-Way ANOVA result of variation on the difference in Teachers' Readiness to Integrate Technology into Teaching-Learning Process based on Years of Experience

Source: Year of Experience	Sum of Squares	Df	Mean Square	F	Sig.	Decision
Between Groups	13.559	4	3.390	2.851*	.026	Significant
Within Groups	172.414	145	1.189			
Total	185.973	149				

Based on years of experience, Table 5 shows the differences in teachers' preparedness to use technology in the teaching-learning process. According to years of experience, teachers' readiness to incorporate technology into the teaching-learning process varies significantly

($F(4,145) = 2.851$, $p = .026 < 0.05$). This suggests that, depending on years of experience, teachers' readiness to use technology in the teaching-learning process varies significantly.

Table 5b: Tukey HSD Post-hoc Analysis of Teachers' Readiness Based on Years of Experience

(I) Years of Experience	(J) Years of Experience	Mean Difference (I-J)	Std. Error	Sig.	Decision
0–5 years	6–10 years	0.176	0.140	.437	Not Significant
0–5 years	11–15 years	0.352*	0.137	.046	Significant
0–5 years	16–20 years	0.418*	0.145	.028	Significant
0–5 years	20 years and above	0.497*	0.151	.017	Significant
6–10 years	11–15 years	0.176	0.139	.421	Not Significant
6–10 years	16–20 years	0.242	0.147	.318	Not Significant
6–10 years	20 years and above	0.321*	0.149	.049	Significant
11–15 years	16–20 years	0.066	0.138	.910	Not Significant
11–15 years	20 years and above	0.145	0.143	.603	Not Significant
16–20 years	20 years and above	0.079	0.148	.882	Not Significant

Mean difference is significant at the 0.05 level

To further determine the specific group differences, a post-hoc analysis using Tukey HSD was conducted. The results, as presented in Table 5a, reveal that significant differences exist between teachers with fewer years of experience (0-5 years) and those with higher years of experience (11-15 years, 16–20 years, and 20 years and above). This indicates that teachers with fewer years of experience show greater readiness to integrate technology than their more experienced counterparts.

Teachers with less experience are more prepared to use technology in the classroom than teachers with more experience, according to this study. This could be because new teachers have more computer experience, having been exposed to them during their training.

The null hypothesis is therefore rejected.

Summary of the Findings

1. This study revealed that secondary school teachers are ready to integrate technology into the teaching-learning process.
2. The study also revealed there is a significant difference in teachers' readiness to integrate technology in the teaching-learning process based on gender.
3. The study also indicated that there is a significant difference in teachers' readiness to integrate technology in the teaching-learning process based on age.

4. The result further shows that there is a significant difference in teachers' readiness to integrate technology into the teaching-learning process based on years of experience.

Discussion of Findings

Research Question one sought to determine the level of teachers' readiness to integrate technology into the teaching-learning process. The findings revealed that secondary school teachers in Olorunda Local Government Area demonstrate a generally positive level of readiness, as indicated by the overall mean score, which is above the criterion value. This suggests that teachers increasingly recognise the importance of digital technologies in enhancing instructional delivery and improving learning outcomes.

This finding may be attributed to the growing exposure of teachers to digital tools and the increasing emphasis on technology-driven pedagogy within the education system. It can be further explained within the framework of the Technological Pedagogical Content Knowledge, which posits that effective technology integration occurs when teachers combine technological, pedagogical, and content knowledge. The observed readiness implies that many teachers are gradually developing this integrated competence. This finding is consistent with Sancar et al. (2021), who reported that teachers' readiness for technology integration is significantly enhanced through continuous professional development and digital competence.

In relation to Hypothesis One, which stated that there is no significant difference in teachers' readiness based on gender, the findings revealed a statistically significant difference ($t = 1.987$, $p = .049$), leading to the rejection of the null hypothesis. The result showed that male teachers demonstrated higher readiness to integrate technology compared to female teachers.

This difference may be explained by variations in technological self-efficacy, access to digital resources, and prior exposure to ICT tools. From the perspective of the Technology Acceptance Model, such differences may stem from variations in perceived ease of use and perceived usefulness of technology across gender groups. This finding aligns with Sofal et al. (2024), who reported gender-based differences in digital competence among secondary school teachers, and Masry-Herzallah (2025), who found that technological self-efficacy significantly influences teachers' adoption of digital technologies.

Hypothesis Two examined differences based on age, the findings indicated a statistically significant difference ($F(4,145) = 2.755$, $p = .030$), and the null hypothesis was therefore rejected. The post-hoc analysis further revealed that younger teachers exhibited higher readiness to integrate technology compared to older teachers.

This pattern suggests that age-related differences in digital exposure and adaptability influence technology integration. Younger teachers, having been trained in more digitally enriched environments, are more likely to possess stronger digital competencies and greater flexibility in adopting new instructional tools. In contrast, older teachers may demonstrate lower confidence or preference for traditional teaching methods. This finding is consistent with the Technology Acceptance Model, which emphasises perceived ease of use as a key determinant of technology

adoption. Supporting this, Falloon (2020) found that younger teachers tend to demonstrate higher digital competence and adaptability.

Hypothesis Three examined differences based on years of teaching experience, the findings revealed a statistically significant difference ($F(4,145) = 2.851, p = .026$), leading to the rejection of the null hypothesis. The post-hoc analysis indicated that teachers with fewer years of experience demonstrated higher readiness to integrate technology compared to their more experienced counterparts.

This finding suggests that teaching experience does not necessarily translate into technological competence, particularly in a rapidly evolving digital environment. Teachers with fewer years of experience are more likely to have been exposed to contemporary digital tools during their training, thereby enhancing their readiness. Within the framework of the Technological Pedagogical Content Knowledge, this highlights the need for continuous updating of technological knowledge alongside pedagogical skills. This finding is in line with Tondeur et al. (2021), who reported that early-career teachers tend to exhibit higher ICT competence due to more recent exposure to technology-integrated training.

Overall, the findings of this study demonstrate that while teachers generally show readiness to integrate technology into the teaching-learning process, this readiness is significantly influenced by demographic factors such as gender, age, and years of teaching experience. These results underscore the importance of targeted professional development programmes that address the specific needs of different teacher groups to promote more inclusive and effective technology integration in secondary schools.

Conclusion

The study examined how teachers' demographic characteristics affected their willingness to use technology in the classroom. Only secondary school teachers in Olorunda Local Government Area of Osun State were included in the study. The findings revealed that teachers generally demonstrate a moderate to high level of readiness to integrate technology into instructional practices. This suggests an increasing awareness among teachers of the relevance of digital technologies in enhancing teaching effectiveness and improving students' learning outcomes.

However, the study further established that this readiness is not uniform across all teachers. Significant differences were observed based on gender, age, and years of teaching experience. Male teachers were found to demonstrate higher levels of readiness compared to their female counterparts, while younger teachers and those with fewer years of teaching experience exhibited greater readiness to integrate technology than older and more experienced teachers.

These findings imply that although the general disposition towards technology integration is positive, disparities exist due to differences in exposure, confidence, and access to technological resources. Therefore, teachers' readiness to integrate technology is not solely dependent on availability of tools, but also on individual and contextual factors that shape their ability and willingness to adopt digital innovations in teaching.

According to Torchia (2023), in order to successfully incorporate digital tools into instruction, teachers must also modify their pedagogical strategies. Effective, technology-enhanced instruction is more likely to be delivered by teachers who match pedagogical approaches with technology applications.

Recommendations

The following recommendations are made, based on the findings of the study;

1. Educational authorities and school administrators should organise continuous and targeted training programmes on technology integration, particularly for older teachers and those with more years of teaching experience, to enhance their digital competence and confidence.
2. Policies and interventions should be designed to address gender disparities in technology use by providing equal access to ICT resources, training opportunities, and support systems for both male and female teachers.
3. Government and school management should ensure the availability of functional ICT facilities, including internet access, computers, and instructional software, to support effective technology integration.
4. School administrators should establish monitoring and support mechanisms, such as mentoring and peer collaboration, to encourage teachers to continuously adopt and effectively use technology in their teaching practices.

References

- Amponsah, S., Boateng, R., & Bediako, Y. (2024). Teachers' confidence and competence in ICT integration: Examining gender differences in technology use. *Cogent Education*, 11(1), 1–15.
- Aondover, E. M., Oyeleye, S. A., & Aliyu, M. A. (2024). New world information and communication order and its changing role in Nigerian Television Authority (NTA) Kano. *Unisia Journal of Social Sciences*. <https://doi.org/10.20885/unisia.vol41.iss1.art2>
- Ayite, D. M., Osei, C. K., & Owusu, E. (2022). Gender dimensions in emerging technologies: Implications for teaching and learning. *International Journal of Educational Technology Research*, 10(2), 45–58.
- Belda-Medina, J., & Calvo-Ferrer, J. R. (2022). Pre-service teachers' knowledge and attitudes toward digital-game-based language learning. *Education Sciences*, 12(3), 182. <https://doi.org/10.3390/educsci12030182>
- Estaji, M., & Zhaleh, K. (2022). The enactment of classroom justice through explicit instruction: Deciphering the changes in English as a foreign language teachers' perceptions and practices. *Frontiers in Psychology*, 13, 846242. <https://doi.org/10.3389/fpsyg.2022.846242>
- Falloon, G. (2020). From digital literacy to digital competence: The teacher digital competency (TDC) framework. *Educational Technology Research and Development*, 68(5), 2449–2472. <https://doi.org/10.1007/s11423-020-09767-4>

- Hennessey, S., D'Angelo, S., McIntyre, N., & Koomar, S. (2022). Technology use for teacher professional development in low- and middle-income countries: A systematic review. *International Journal of Educational Research Open*, 3, 100174. <https://doi.org/10.1016/j.ijedro.2022.100174>
- Hermanto, Y. B., & Srimulyani, V. A. (2021). The challenges of online learning during the COVID-19 pandemic. *Jurnal Pendidikan dan Pengajaran*, 54(1), 46–57.
- Joseph, G. V., Thomas, K. A., & Nero, A. (2021). Impact of technology readiness and technostress on teacher engagement in higher secondary schools. *Digital Education Review*, 40, 51–65. <https://doi.org/10.1344/der.2021.40.51-65>
- Makhlouf, K., & Bensafi, Z. (2021). An exploration of factors influencing teachers' attitudes toward the use of information and communication technology (ICT) in classroom practice. *Advances in Language and Literary Studies*, 12(2), 37–45. <https://doi.org/10.7575/aialc.all.v.12n.2p.37>
- Masalimova, A. R., Erdyneeva, K. G., Kislyakov, A. S., Sizova, Z. M., Kalashnikova, E., & Khairullina, E. R. (2022). Validation of the scale on pre-service teachers' digital competence. *Contemporary Educational Technology*, 14(4), ep390. <https://doi.org/10.30935/cedtech/12334>
- Masry-Herzallah, A. (2025). TPACK, technological self-efficacy, gender, and online teaching effectiveness. *Humanities and Social Sciences Communications*, 12, 227. <https://doi.org/10.1038/s41599-025-04546-z>
- Merriam-Webster. (2016). Demographic variable. In *Merriam-Webster's dictionary*.
- Momdjian, L., Manegre, M., & Gutiérrez-Colón, M. (2024). A comparison of perceptions of digital competences of schoolteachers and school leaders in Lebanon. *Social Sciences & Humanities Open*, 10, 100937. <https://doi.org/10.1016/j.ssaho.2024.100937>
- OECD. (2021). *21st-century readers: Developing literacy skills in a digital world*. OECD Publishing.
- Qureshi, M., Khan, N., Raza, H., Imran, A., & Ismail, F. (2021). Digital technologies in education 4.0: Does it enhance the effectiveness of learning? <https://ir.unikl.edu.my/xmlui/handle/123456789/25661>
- Sancar, R., Atal, D., & Deryakulu, D. (2021). Teacher professional development in the digital age: A systematic review. *Technology, Knowledge and Learning*, 26(4), 877–905. <https://doi.org/10.1007/s10758-020-09477-6>
- Sancar, R., Atal, D., & Deryakulu, D. (2021). A new framework for teachers' professional development. *Teaching and Teacher Education*, 101, 103305. <https://doi.org/10.1016/j.tate.2021.103305>
- Siddiq, F., & Scherer, R. (2016). Is there a gender gap? A meta-analysis of gender differences in students' ICT literacy. *Educational Research Review*, 18, 1–17. <https://doi.org/10.1016/j.edurev.2016.03.007>
- Sofal, F. A., Ansari, M. I. Y., Akram, M., & Pradhan, M. (2024). Gender differences in professional and digital competencies among secondary school teachers. *i-manager's*

- Journal on School Educational Technology*, 20(2), 1–11.
<https://doi.org/10.26634/jsch.20.2.21160>
- Tondeur, J., Scherer, R., Siddiq, F., & Baran, E. (2021). A comprehensive analysis of preservice teachers' ICT competencies. *Computers & Education*, 162, 104080.
<https://doi.org/10.1016/j.compedu.2020.104080>
- Torchia, R. (2023). Technology in the classroom & the benefits for K–12 schools.
<https://edtechmagazine.com>
- UNESCO. (2023). *Global education monitoring report 2023: Technology in education*. UNESCO Publishing.
- World Bank. (2020). *Realizing the future of learning: From learning poverty to learning for everyone, everywhere*. World Bank.
- Zulueta, M. C. E., Sagun, A. T., Hero, J. L., Gloria, D. S., Tongol, J. C. L., Cruz, A. C. D., Cajurao, G. V., & Cabrera, W. C. (2021). Mastering innovations in the lens of ICT competence and practices of 21st century Filipino teachers. <https://ejournals.ph/article.php?id=16787>